AI Clinical Report
Summarization Assistant

Team 1

The business problem

Report review is slow, inconsistent, and difficult to audit.

Today

1. Read every report page by page.
2. Check units and ranges by eye.
3. Reconstruct the source chain later.

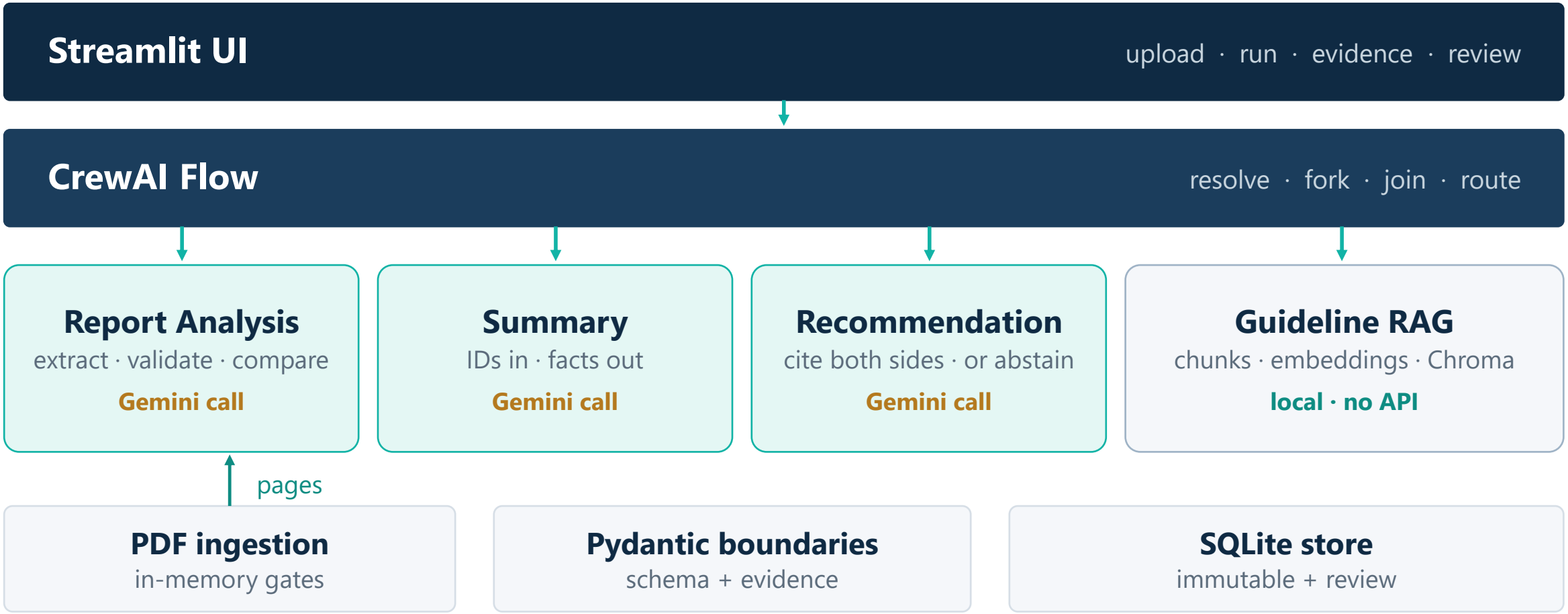
Our solution

1. **Validate source**
   page + exact excerpt
2. **Retrieve + summarise**
   local guidance, deterministic facts
3. **Recommend + review**
   cite or abstain; a human decides

A traceable draft, not an autonomous diagnosis.

Architecture

Components, boundaries, and three bounded agents.



Teal = agent · amber = Gemini boundary · grey = deterministic support

Ingestion + analysis

From PDF bytes to source-verified findings.

PDF intake

Size & type
≤ 10 MiB · %PDF

Parse
pypdf · page text

Identity
SHA-256 · in memory

No text
OCR warning only

No PDF bytes written to disk.

Extract
findings → **Validate**
page + excerpt → **Compare**
Decimal

Verified synthetic example

110 mg/dL vs 70–99 mg/dL **above_range**

SEPARATED OUTPUTS

Patient | **Narrative** | **Control** | **Warnings**

The model proposes. Code checks.

Missing range → not assessed · unit mismatch → review

Guideline RAG

Local retrieval with exact page provenance.



Every chunk knows

document · page · hash

Index fingerprint

SHA-256 config + document hashes

EXACT SYNTHETIC PASSAGE · PAGE 2

"Do not infer an interval."

Missing reference-range review

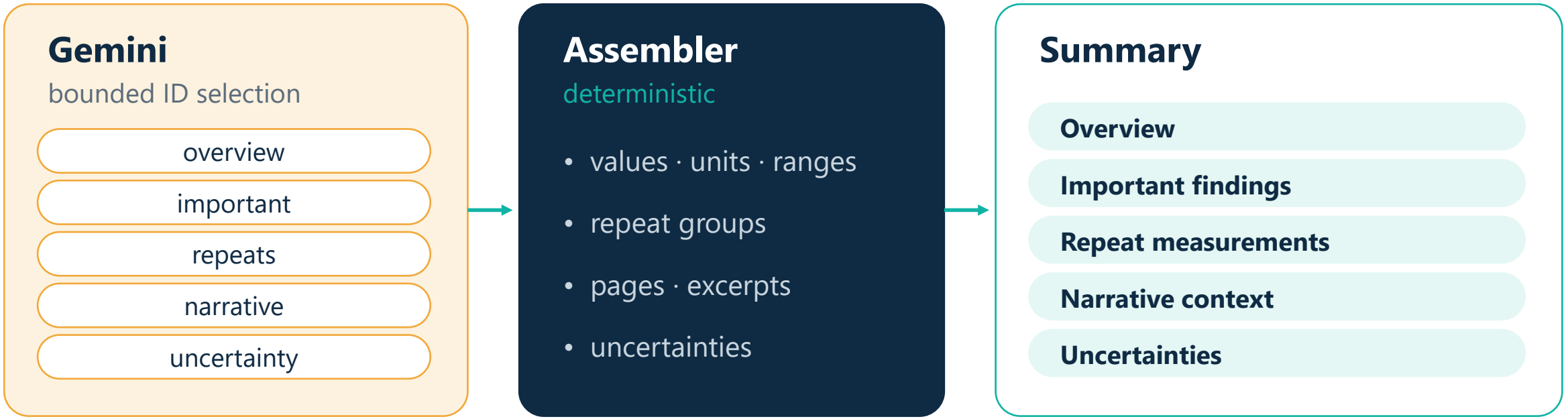
Similarity ranks relevance.

It is not confidence.

Local embeddings · persistent Chroma · no Gemini call in retrieval

Summary

Gemini selects IDs; deterministic code writes the facts.



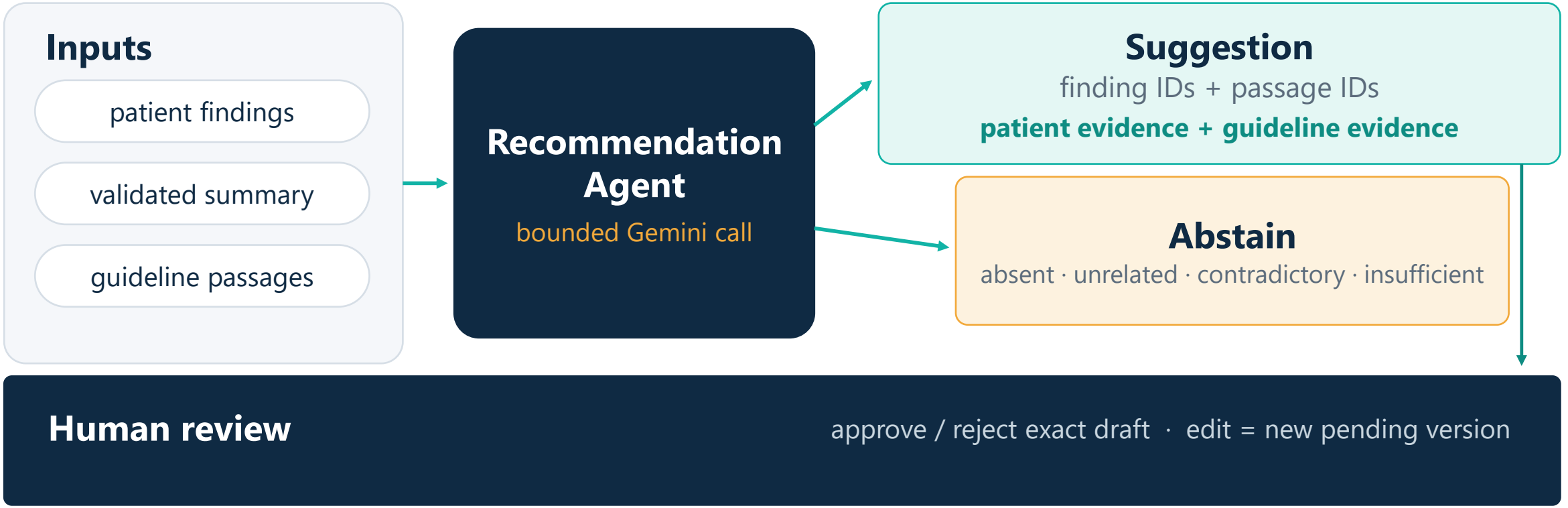
IDs in. Facts out.

The model never writes the numeric fact.

Repeat groups: matching analyte + unit; no invented chronology.

Recommendation + human review

Every suggestion cites both sides — or declines.

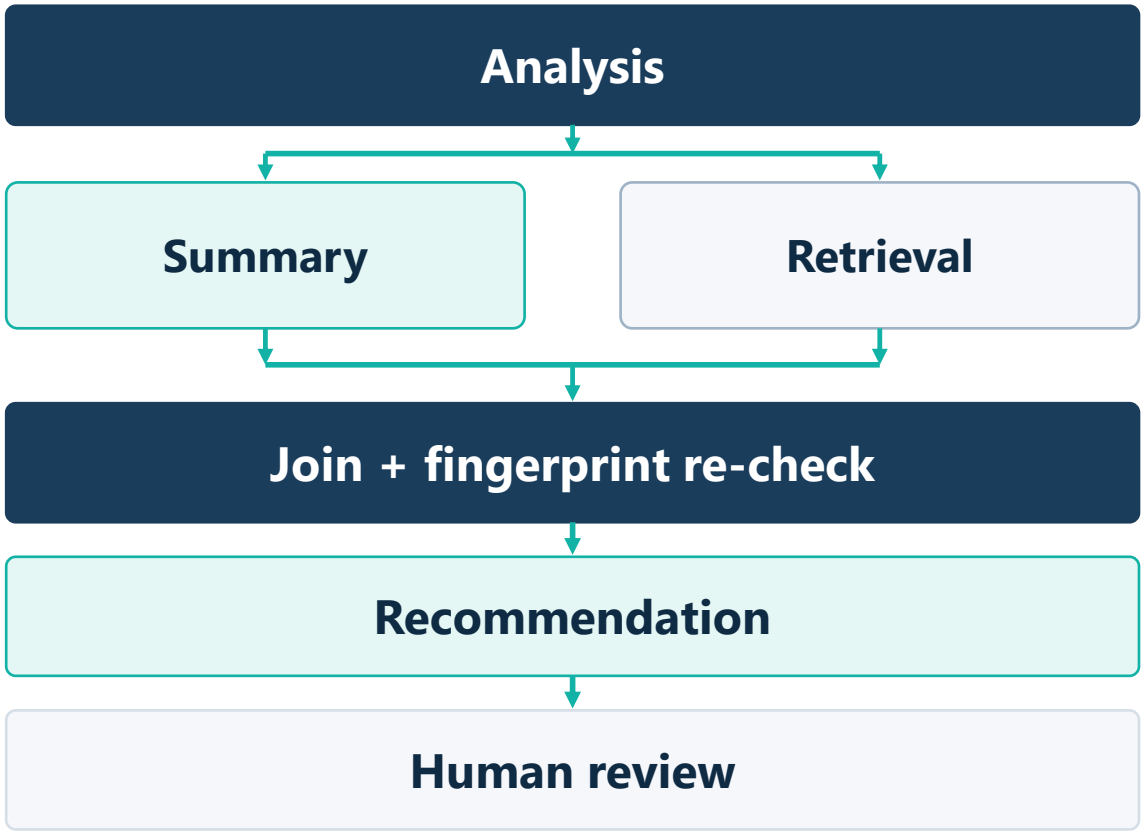


No diagnosis · no medication changes · no doses · no urgency

Approval is a review decision, not clinical certification.

Orchestration + persistence

Execution order, parallel branches, and exact reuse.



SQLite
immutable local results

- immutable records
- exact compatibility keys
- revalidate on load

No new Gemini call for a reused stage.
Exact keys + validated load.

reuse · generate · retrieve · abstain · await review · approved · rejected

A failed branch preserves every saved upstream stage.

Validation and limitations

Offline evidence is explicit; live behavior is checked separately.

173 **tests passed**

1 optional test skipped; Ruff + lockfile clean

0 **API calls in mocked demo**

No network, key, or model download

1 **manual live smoke request**

Passed separately from offline tests

Limitation: not a medical device

Synthetic educational demo only. No diagnosis, treatment, or clinical advice.

Limitation: PDFs and similarity

Scanned PDFs need OCR, which is out of scope. Similarity is not confidence.

Show the source. Decline when unsure. Let a human decide.